Status	Finished
Started	Friday, 16 October 2026, 8:17 AM
Completed	Friday, 16 October 2026, 8:27 AM
Duration	9 mins 24 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. sort()
- c. arrange()
- d. sorted()

The correct answer is: sorted()

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Dividing the input into parts, solving each part, and combining the solutions
- c. Sorting data in a single pass
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Arranging data in a particular format
c. Deleting data from a list
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting decreases the efficiency of selection operations
b. Sorting makes it harder to search for items
c. Sorting is rarely used in programming
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Linear Search
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It is less efficient than custom sorting algorithms
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **9**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Using the divide-and-conquer approach
c. Being the most efficient sorting algorithm
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It divides the list into two halves, sorts each half, and then merges them
c. It compares adjacent elements and swaps them if necessary
d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is difficult to implement
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Binary Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are combined to form a single sorted list
b. They are discarded
c. They are split again into smaller sublists
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Finding a specific element in a list
c. Removing duplicates from a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **15**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the largest element to its correct position
c. Bubbling up the smallest element
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position